

Question Bank

Unit – 1

1. **Justify the term cloud computing and provide a case study of migrating from physical server to cloud.**

Cloud computing is a model for delivering computing resources such as servers, storage, databases, networking, and software over the Internet on demand. It allows users to access and use computing services without owning or managing the underlying physical infrastructure. Resources are provided by a cloud service provider and are accessed through the network as required. The term "cloud" represents the Internet or a network of remote servers that provide computing services. In network diagrams, the Internet is often illustrated as a cloud symbol to hide the complexity of the underlying infrastructure.

- **Remote Resource Access**
Computing resources are not located on the user's local machine. They are hosted in remote data centers and accessed through the Internet.
- **Shared Infrastructure**
Multiple users share the same physical infrastructure. Virtualization technology divides physical servers into multiple virtual machines that serve different users.
- **On-Demand Service**
Users can obtain computing resources whenever required without installing hardware.
- **Elastic Resource Allocation**
Resources such as CPU, memory, and storage can be increased or decreased depending on the workload.

Service-Based Model

Cloud computing provides services such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

Because computing services are delivered through remote infrastructure represented as a cloud in network diagrams, the term "cloud computing" is used.

Case Study: Migration from Physical Server to Cloud

A medium-sized e-commerce company initially hosted its website and database on an on-premise physical server located in its office data center. The infrastructure consisted of one web server, one database server, and local storage systems.

Problems with Physical Servers

The company faced several operational challenges:

- * Limited computing capacity during high traffic periods.
- * High cost of hardware maintenance and upgrades.
- * Risk of service interruption due to hardware failure.
- * Difficulty in performing backup and disaster recovery.

These limitations led the company to migrate its infrastructure to a cloud environment.

Migration Process

1. Assessment of Existing Infrastructure

The organization evaluated the following components:

- * Web application server
- * Database server
- * Storage requirements
- * Network configuration

The goal was to determine the resources needed in the cloud environment.

2. Selection of Cloud Service Model

The company selected Infrastructure as a Service (IaaS) so that it could deploy virtual machines similar to its existing physical servers.

3. Data and Application Migration

The migration steps included:

- * Creating virtual machines in the cloud.
- * Transferring application code to cloud servers.
- * Migrating the database to cloud storage.
- * Configuring network access and security rules.

4. Testing and Deployment

After migration, the system was tested for:

- * Application performance
- * Data integrity
- * Network connectivity

2. Discuss the term virtualisation and provide a justification of what is a virtualisation, purpose of virtualisation and benefits of virtualisation

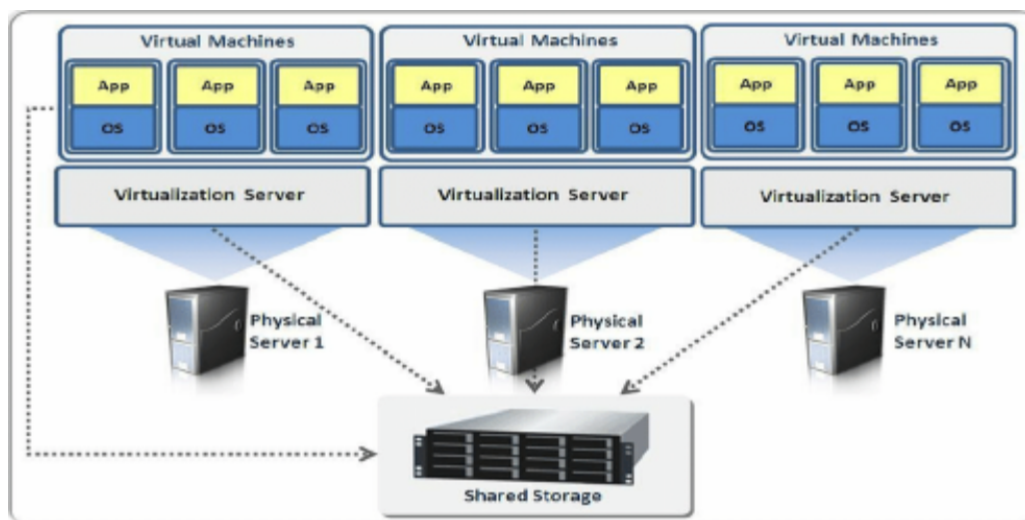
Virtualization represents a technology platform used for the creation of virtual instances of IT resources. A layer of virtualization software allows physical IT resources to provide multiple virtual images of themselves so that underlying processing capabilities can be shared by multiple users. Prior to the advent of virtualization technologies, software was limited to residing on and being coupled with static hardware environments. The virtualization process severs this software hardware dependency, as hardware requirements can be simulated by emulation software running in virtualized environments. Virtualization is justified because it abstracts physical IT resources and removes the direct dependency between software and hardware.

Through virtualization:

- Physical IT resources can provide multiple virtual images of themselves.
- Processing capabilities can be shared by multiple users.
- Software is no longer tightly coupled with specific hardware environments.
- Established virtualization technologies have inspired many core cloud computing features.

The purpose of virtualization includes:

- **Creation of Virtual Instances**
To create virtual instances of servers, storage, and other IT resources.
Resource Pooling and Multitenancy
Multitenancy models frequently rely on virtualization technologies.
IT resources can be dynamically assigned and reassigned according to cloud service consumer demands.
Support for Elasticity
Elasticity is the automated ability of a cloud to scale IT resources as required.
Benefits of Virtualization
- Based on the Goals and Benefits section in the PDF, virtualization supports the following benefits in cloud environments:
Reduced Investments and Proportional Costs
Reduction or elimination of up-front IT investments and operational costs proportional to measured usage.
Increased Scalability
IT resources can be dynamically allocated and released according to demand.
Increased Availability and Reliability
Cloud environments provide support for increasing availability and reliability through resilient IT resources and failover mechanisms.



3. What is business influencer or drivers in establishing cloud compute. With an example justify Business Drivers for Establishing Cloud Computing

Business drivers in cloud computing are the economic, operational, and strategic factors that motivate an organization to adopt cloud-based infrastructure and services instead of maintaining traditional on-premise IT systems.

These drivers focus on cost efficiency, flexibility, faster deployment, and risk reduction.

1. Financial Drivers

a) Reduction of Capital Expenditure

Traditional IT infrastructure requires high initial investment in servers, storage devices, networking equipment, and data centers.

Cloud computing follows a pay-per-use model, which converts Capital Expenditure into Operational Expenditure.

b) Lower Total Cost of Ownership

The cloud service provider manages hardware maintenance, upgrades, cooling, and physical security. This reduces long-term operational and maintenance costs for the organization.

2. Operational Drivers

a) Rapid Provisioning

Cloud platforms allow computing resources such as virtual machines and storage to be created within minutes.

This reduces deployment time for new applications and services.

b) Elasticity

Cloud systems automatically increase or decrease resources based on demand.

This prevents over-provisioning and under-utilization of hardware.

3. Strategic Drivers

a) Business Continuity

Cloud providers maintain data replication across multiple geographic regions.

This ensures service availability during hardware failure, natural disaster, or power outage.

b) Global Accessibility

Applications hosted in the cloud can be accessed from any location using the internet, enabling global operations and remote work.

Justification with Example

Consider an e-commerce company that experiences heavy traffic during festival sales.

In a traditional data center:

The company must purchase enough servers to handle peak traffic.

These servers remain idle during normal days.

This results in unnecessary capital investment.

In a cloud environment:

During peak sales, additional virtual machines are automatically provisioned.

After the sale ends, extra resources are released.

The company pays only for the duration of high usage.

Thus, cloud computing aligns IT cost with business demand and reduces financial risk.

4. What are basic concepts of cloud. Justify the terminologies a. IT resources b. On premises c. Cloud service providers d. Scaling

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

A. IT Resources

In the context of cloud computing, an IT resource is any hardware or software-based artifact used to physicalize or virtualize a computing environment. These are the building blocks of any digital service.

Physical IT Resources: These include physical servers (compute), hard disks or flash arrays (storage), and switches or routers (networking).

Virtual IT Resources: These are abstracted versions of physical hardware, such as Virtual Machines (VMs), virtual storage volumes, or software-defined networks.

Software Resources: This includes operating systems, database management systems (DBMS), and application frameworks provided as a service.

B. On-premises

On-premises (often shortened to "on-prem") refers to the traditional model where an organization maintains its own physical data center. In this model, the organization is responsible for the entire stack of IT infrastructure.

Ownership: The organization owns the physical building, the hardware, and the networking equipment.

Management: Internal IT staff must handle hardware maintenance, cooling, power redundancy, and physical security.

Capital Expenditure (CapEx): Significant upfront investment is required to purchase and install the infrastructure before it can be used.

C. Cloud Service Providers (CSPs)

A Cloud Service Provider is an entity that owns and operates large-scale data centers and offers IT resources to consumers over a network, typically the Internet.

Role: The CSP manages the underlying physical infrastructure, virtualization layer, and often the middleware or application software.

Service Delivery: Resources are delivered via service models such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

Economic Model: CSPs typically use a "pay-as-you-go" or utility-based pricing model, converting IT costs from Capital Expenditure (CapEx) to Operational Expenditure (OpEx).

D. Scaling

Scaling refers to the ability of an IT resource to handle increasing or decreasing demand by adjusting its capacity. This is a primary driver for cloud adoption.

Vertical Scaling (Scaling Up/Down): This involves changing the capacity of an existing resource, such as upgrading a server with more RAM or a faster CPU. It is limited by the maximum capacity of a single physical machine.

Horizontal Scaling (Scaling In/Out): This involves adding or removing resource instances. For example, adding more web servers to a cluster to handle a spike in traffic. This provides higher availability and is theoretically limitless in a cloud environment.

5. For a service to be deployed in a cloud it requires a dedicated role to be played by various individual. With a case study or an example Justify various contributors and roles in stating the services

For a service to be deployed in a cloud environment, different individuals and organizations must perform specific roles. Clear distribution of responsibilities ensures proper functioning, security, and

reliability of the service.

Major Contributors and Their Roles:

- **Cloud Service Provider**
The provider supplies infrastructure such as servers, storage, and networking through data centers. It ensures availability, maintenance, and basic security of the cloud platform.
- **Cloud Consumer**
The consumer is the organization or user that utilizes the cloud service. The consumer manages application settings, user access, and data according to the selected service model.
- **Cloud Service Developer**
The developer designs and builds the cloud application. Responsibilities include coding, testing, database integration, and ensuring scalability.
- **Cloud Administrator**
The administrator monitors performance, allocates resources, manages backups, and handles operational issues.
- **Security Manager and Auditor**
The security manager implements data protection measures such as encryption and authentication. The auditor verifies compliance with standards and policies. Example

Consider an online shopping website deployed on a public cloud. The cloud service provider offers virtual servers and storage. The developer creates the website and payment system. The administrator monitors traffic and scales resources during peak sales. The security manager protects customer data, while the auditor checks financial compliance. The retail company, as the consumer, manages products and customer information. Successful cloud deployment requires coordination among providers, developers, administrators, security personnel, and consumers. Clearly defined roles ensure secure, scalable, and efficient cloud services.

6. Discuss the goals and benefits of migrating services from regular physical server to cloud services

Cloud migration is the transition of applications, data, and IT processes from on-premises, physical hardware to a virtualized environment managed by a Cloud Service Provider (CSP).

Goals of Migrating to Cloud Services

1. Cost Reduction

The primary goal is to reduce capital expenditure on hardware, networking devices, and data center infrastructure. Organizations aim to convert capital expenses into operational expenses by paying only for the resources used.

2. On-Demand Resource Provisioning

Cloud environments allow resources such as virtual machines, storage, and networking to be provisioned as needed. The goal is to avoid underutilization or overprovisioning of physical servers.

3. High Availability and Reliability

Migration aims to improve service uptime. Cloud providers offer distributed infrastructure and redundancy to reduce service interruption.

4. Improved Performance

Cloud platforms provide access to modern hardware, high-speed networking, and distributed computing capabilities, improving overall system performance.

5. Business Continuity and Disaster Recovery

A key goal is to ensure data backup and quick recovery in case of system failure, natural disasters, or cyberattacks.

6. Simplified Infrastructure Management

Cloud services reduce the need for managing physical hardware, patching servers, and maintaining data centers.

Benefits of Migrating to Cloud Services

1. Cost Efficiency

No need for purchasing and maintaining physical servers
Reduced electricity and cooling costs
Pay-as-you-use pricing model

2. Elasticity

Resources can be increased or decreased based on workload demand
Suitable for applications with variable traffic

3. Scalability

Supports horizontal and vertical scaling
Enables growth without major infrastructure changes

4. High Availability

Data replication across multiple geographic locations
Fault tolerance through distributed systems

5. Enhanced Security

Cloud providers implement security measures such as encryption, firewalls, and identity management
Regular security updates and compliance certifications

6. Faster Deployment

Rapid provisioning of servers and storage
Reduced time required to launch applications

7. Automatic Updates and Maintenance

Infrastructure maintenance handled by cloud provider
Reduced administrative workload

8. Global Accessibility

Services accessible from different geographic locations

Supports remote work and distributed teams

Example

An organization running an e-commerce website on physical servers must purchase additional hardware during festive seasons to handle high traffic. After migration to cloud services, the organization can dynamically increase computing resources during peak demand and reduce them afterward, thereby lowering cost and improving performance.

7. Is cloud computing a risk free facility. Justify the opinion and provide a detailed overview of various risks and challenges associated with the cloud

Cloud computing is not a risk-free environment. Although cloud providers implement security controls, the shared and distributed nature of cloud infrastructure introduces inherent vulnerabilities. Data and applications are hosted on remote systems managed by third-party providers, reducing direct user control. As a result, risks related to security, privacy, availability, and compliance may arise.

Various Risks and Challenges in Cloud Computing

1. Data Security and Privacy Risk

Unauthorized access, data breaches, or insider attacks may expose sensitive information stored in the cloud.

2. Data Loss

Accidental deletion, cyber attacks, hardware failure, or weak backup mechanisms can result in permanent data loss.

3. Service Downtime

Cloud services may become unavailable due to server failures, network issues, or cyber attacks, affecting business operations.

4. Network Dependency

Cloud access depends on internet connectivity; poor bandwidth or network failure disrupts services.

5. Multi-Tenancy Risk

Multiple users share the same physical resources, increasing the risk of data leakage between tenants.

6. Virtualization Vulnerabilities

Security flaws in hypervisors may allow attackers to access multiple virtual machines.

7. Compliance and Legal Issues

Data stored across different countries may violate data protection laws and regulatory requirements.

8. Who is a cloud service provider and a cloud service owner and a cloud service consumer. With an example explain the individual.

Cloud Service Provider (CSP):

Definition: A Cloud Service Provider is an entity or organization responsible for making cloud services available to interested parties.

Explanation: The CSP builds, manages, and maintains the necessary IT infrastructure, including servers, storage, networking, power, cooling, and the virtualization layer. They are responsible for the physical security of the data center and ensuring that the service meets the promised Service Level Agreements (SLAs), focusing on availability, performance, and scalability of the raw resources.

Cloud Service Owner (CSO):

Definition: A Cloud Service Owner is a person or organization that holds the legal ownership of a specific cloud service.

Explanation: The CSO deploys, configures, and maintains the specific software application, platform, or service that utilizes the underlying infrastructure. The CSO is responsible for defining the service specifications, setting the access policies, and managing the runtime environment of the service. Depending on the service model, the CSO might be a third party renting infrastructure from a CSP, or the CSP themselves (e.g., in a SaaS model).

Cloud Service Consumer (CSC):

Definition: A Cloud Service Consumer is an organization or human individual that subscribes to and utilizes the services offered by a cloud service owner or provider.

Explanation: The CSC is the end user of the cloud service. Their interaction with the service occurs through various interfaces such as web browsers, mobile applications, or Application Programming Interfaces (APIs). The consumer does not manage the underlying cloud infrastructure or the core configuration of the application, focusing solely on using the service's functionality.

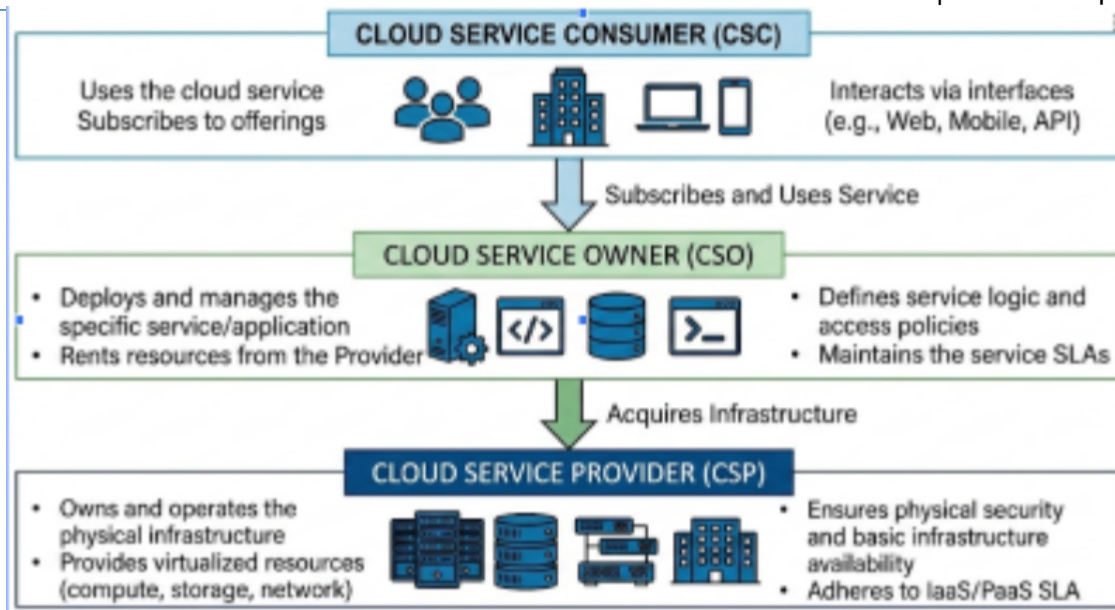
Integrated Example:

Consider a SaaS-based Customer Relationship Management (CRM) application.

1) Cloud Service Provider (CSP): Company A owns data centers globally and provides virtualized compute, storage, and networking resources. Company B rents this infrastructure to run their application. Company A is the CSP.

2) Cloud Service Owner (CSO): Company B developed the CRM application, deployed it on Company A's infrastructure, and maintains the software code, application security, and subscription models. Company B is the CSO.

3) Cloud Service Consumer (CSC): Company C is a business that subscribes to the CRM application. Its sales employees use the application daily via a web interface to log client interactions. Company C (and its employees) are the CSCs.



9. What is the role of cloud auditor and cloud broker. Justify

Role of Cloud Auditor and Cloud Broker in Cloud Computing

1. Cloud Auditor

A Cloud Auditor is an independent entity that evaluates cloud services to ensure compliance with security policies, performance standards, regulatory requirements, and Service Level Agreements (SLA).

- Functions of a Cloud Auditor
- Security Assessment
- Examines data protection mechanisms.
- Verifies encryption methods and access control policies.
- Compliance Verification
- Ensures adherence to legal and industry standards.
- Checks regulatory compliance such as data protection laws.
- Performance Evaluation
- Validates service performance against agreed SLA terms.
- Assesses availability, reliability, and response time.
- Risk Assessment
- Identifies vulnerabilities in cloud infrastructure.
- Evaluates potential operational risks.
- Justification of Role
- Ensures transparency between cloud provider and customer.
- Builds trust by validating provider claims.
- Protects customer data through independent verification.
- Confirms that SLA commitments are fulfilled.

Thus, a cloud auditor plays a critical role in maintaining accountability and trust in cloud environments.

2. Cloud Broker

A Cloud Broker is an intermediary entity that manages the use, performance, and delivery of cloud services between cloud providers and cloud consumers.

Functions of a Cloud Broker

- Service Intermediation
- Enhances services by adding value such as security or monitoring.
- Service Aggregation
- Combines multiple cloud services into a single unified service.
- Ensures interoperability among different providers.
- Service Arbitrage
- Selects cloud services from various providers based on cost or performance.
- Cost Management
- Compares pricing models.
- Helps customers select cost-effective solutions.

Justification of Role

- Simplifies multi-cloud management.
- Reduces complexity in vendor selection.
- Improves service efficiency through proper coordination.
- Assists organizations in meeting business and technical requirements.

Therefore, a cloud broker acts as a mediator that ensures efficient service selection and management.

10. What are boundaries defined for in the cloud. With a neat diagram explain the various boundaries and their influencers for a smoother execution of cloud services

Boundaries refer to the defined limits that separate security domains, responsibilities, network areas, and trust levels within a cloud environment. These boundaries help in protecting resources, managing access, ensuring compliance, and enabling the smooth execution of cloud services.

Cloud environments are distributed and shared, so clearly defined boundaries are essential to avoid security risks, performance issues, and responsibility confusion.

1. Trust Boundary

A trust boundary defines the point where the level of trust changes, such as between internet users and internal cloud systems. It ensures that only authenticated and authorized users can access cloud resources. Influenced by Identity and Access Management (IAM), authentication mechanisms, encryption, and Zero Trust policies.

2. Security Boundary

A security boundary protects cloud infrastructure and data from threats. It includes firewalls, Virtual Private Clouds (VPCs), security groups, and intrusion detection systems.

It is influenced by security policies, encryption standards, and monitoring tools.

3. Network Boundary

A network boundary controls communication between internal and external networks. It manages traffic using subnets, VPNs, load balancers, routing tables, and DNS configurations.

It ensures safe and efficient data transmission.

4. Responsibility Boundary (Shared Responsibility Model)

This boundary defines the division of responsibilities between the cloud provider and the customer.

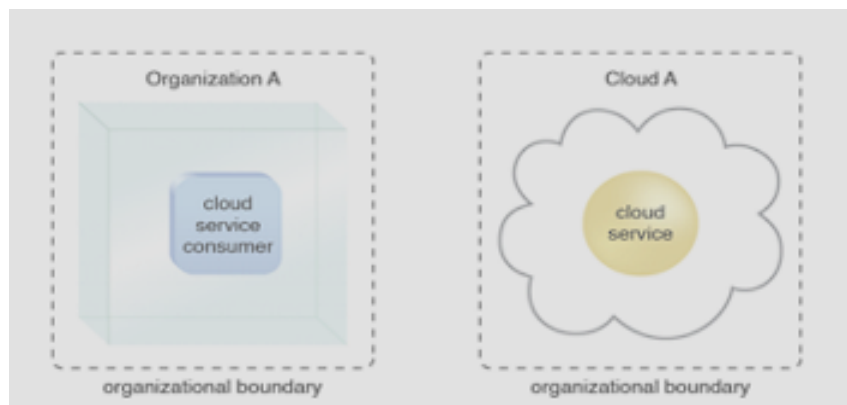
For example, in Amazon Web Services, the provider manages hardware, networking, and data centers, while the customer manages applications, data, and user access.

The responsibility division depends on the service model such as IaaS, PaaS, or SaaS.

5. Compliance Boundary

The compliance boundary ensures that cloud services follow legal and regulatory requirements such as data protection and residency laws.

It is influenced by government regulations, industry standards, and audit requirements.



11. What are cloud characteristics. Explain with an example

Cloud Characteristics

Definition : Cloud computing characteristics are the fundamental features that define the nature and operation of cloud services. These characteristics distinguish cloud computing from traditional computing systems. According to standard cloud models, the following five essential characteristics define cloud computing.

1. On-Demand Self-Service

Explanation

Users can provision computing resources such as servers, storage, and applications automatically without requiring human interaction from the service provider.

Resources can be allocated whenever needed through a web interface or API.

Example

A developer can create a virtual machine instantly using a cloud portal without contacting the service provider.

2. Broad Network Access

Explanation

Cloud services are accessible over the network using standard mechanisms such as web browsers, mobile devices, laptops, and workstations.

This ensures that services are available from different locations and platforms.

Example

Users can access cloud storage services like Google Drive from a smartphone, laptop, or tablet using the internet.

3. Resource Pooling

Explanation

The cloud provider pools computing resources such as storage, memory, and processing power to serve multiple customers using a multi-tenant model.

Physical and virtual resources are dynamically assigned according to user demand.

Example

Multiple organizations share the same physical server infrastructure in a data center, but each organization's data remains logically separated.

4. Rapid Elasticity

Explanation

Cloud resources can be quickly scaled up or scaled down based on demand.

To the user, resources appear unlimited and can be adjusted at any time.

Example

During a festival sale, an e-commerce company increases server capacity to handle high traffic and reduces it after the sale ends.

5. Measured Service

Explanation

Cloud systems automatically monitor and control resource usage.

Users are charged based on actual consumption such as CPU hours, storage space, or bandwidth.

Example

A company pays only for the number of virtual machine hours used in a month instead of purchasing physical hardware.

12. What is infrastructure as a service (IAAS). With a case study explain the architecture and the characteristics of IAAS.

Infrastructure as a Service (IaaS) is a cloud service model in which computing infrastructure such as virtual machines, storage, and networking resources are provided to users over the internet on a pay per use basis. The cloud provider manages the physical hardware, data centers, and virtualization layer, while the user manages the operating system, applications, and data.

IaaS is one of the three primary cloud service models, along with Platform as a Service and Software as a Service.

Architecture of IaaS:

The architecture of IaaS can be understood as a layered structure.

1. Physical Infrastructure Layer:

This is the bottom layer and consists of:

Physical servers

Storage devices

Networking equipment

Data center facilities

These resources are owned and maintained by the cloud provider.

2. Virtualization Layer

This layer uses a hypervisor to create multiple virtual machines on a single physical server.

Type 1 hypervisors such as VMware ESXi run directly on hardware.

Type 2 hypervisors such as Oracle VM VirtualBox run on top of a host operating system.

Virtualization enables resource sharing and isolation among users.

3. Virtual Resource Layer

This layer includes:

Virtual machines

Virtual storage

Virtual networks

Users can configure CPU, memory, storage size, and network settings as required.

4. Management and Orchestration Layer

This layer provides:

Resource provisioning

Monitoring

Billing

Load balancing

Security management

Users access these services through web portals or APIs.

Case Study: Amazon EC2

Amazon Elastic Compute Cloud, commonly known as Amazon EC2, is an example of IaaS provided by Amazon Web Services.

Architecture of Amazon EC2

Data Centers and Regions

AWS operates multiple regions and availability zones worldwide.

Physical Servers

High performance servers are installed in data centers.

Hypervisor

AWS uses virtualization technology to create multiple EC2 instances from physical servers.

EC2 Instances

Users select instance types based on required CPU, memory, and storage.

Storage Services

Elastic Block Store for persistent storage

Object storage through Amazon S3

Networking

Virtual Private Cloud enables users to create isolated virtual networks.

Management

Users manage resources through AWS Management Console or APIs.

Working Process

A user logs into AWS.

Selects an instance type.

Chooses operating system such as Linux or Windows.

Launches the virtual machine.

Deploys applications on the instance.

The provider manages hardware maintenance, power supply, and network infrastructure.

Characteristics of IaaS

1. On Demand Self Service

Users can provision computing resources automatically without human intervention from the provider.

2. Resource Pooling

Physical resources are shared among multiple users using virtualization.

3. Rapid Provisioning

Virtual machines can be created or removed quickly.

4. Pay Per Use Model

Users are charged based on actual resource consumption such as compute hours and storage usage.

5. Scalability

Users can increase or decrease resources based on workload requirements.

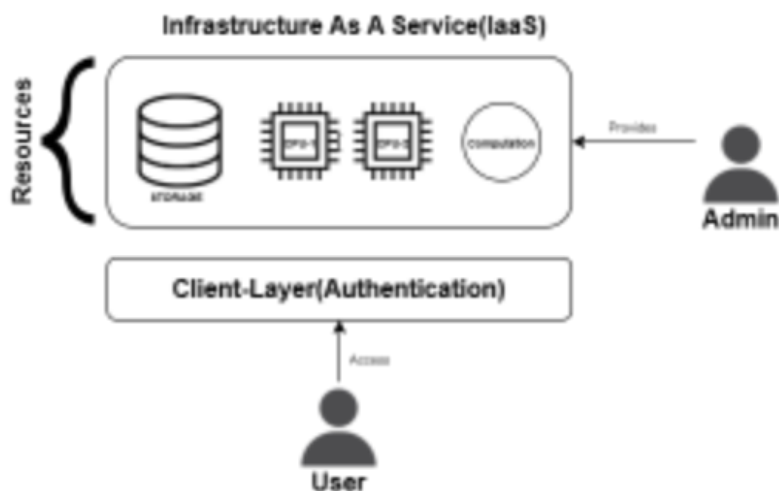
6. Virtualization Based Infrastructure

All computing resources are delivered in virtualized form.

7. Network Access

Resources are accessible over the internet through standard mechanisms.

Infrastructure as a Service provides virtualized computing resources over the internet. It allows users to control operating systems and applications while the provider manages hardware and infrastructure. Services such as Amazon EC2 demonstrate the architecture and operational characteristics of IaaS in real world cloud environments.



13. What is software as a service(SAAS). With a case study explain the architecture and the characteristics of SAAS.

Software as a Service (SaaS) is a cloud service model in which application software is hosted and managed by a cloud provider and delivered to users over the Internet. Users access the software through a web browser or thin client. The provider manages the infrastructure, operating system, application updates, security, and maintenance, while the user only uses the application.

Architecture of SaaS

The architecture of SaaS is generally organized into layered components.

1. Infrastructure Layer

- Consists of physical servers, storage systems, and networking components.
- Uses virtualization to create virtual machines.
- Fully managed by the cloud provider.

2. Platform Layer

- Includes operating systems, middleware, and runtime environments.
- Provides database management systems and application servers.
- Supports the execution of SaaS applications.
- Managed by the cloud provider.

3. Application Layer

- Contains the software applications delivered to end users.
- Supports multi-tenant architecture, where multiple users share the same application instance.
- Handles user authentication, business logic, and data processing.

- On-Demand Access – Applications are accessed through the Internet without installation.

- Multi-Tenancy – A single application instance serves multiple customers with logical data separation.
- Subscription-Based Model – Users pay monthly or annually based on number of users or usage.
- Centralized Management – Maintenance, updates, and security patches are handled by the provider.
- Automatic Updates – Users always access the latest version of the application.
- High Availability – Services are available continuously through cloud data centers.
- Device Independence – Applications can be accessed from any device with Internet connectivity.

Case Study: Salesforce CRM

Salesforce CRM is a SaaS-based Customer Relationship Management application delivered through the Internet.

- Users log in through a web browser.
- The application manages customer records, sales data, and reports.
- Data is stored in centralized cloud databases.
- The cloud provider manages servers, updates, security, and scaling.
- Multiple organizations use the same software platform with separate data storage.

14. What is platform as a service(PAAS). With a case study explain the architecture and the characteristics of PAAS.

Platform as a Service (PaaS) is a cloud service model that provides a development and deployment environment over the internet. It includes operating systems, runtime environments, middleware, databases, and development tools.

The cloud provider manages the infrastructure such as servers, storage, networking, and virtualization, while the user develops and manages applications.

Architecture of PaaS

The architecture of PaaS is typically organized into layered components.

1. Infrastructure Layer

- Consists of physical servers, storage devices, and networking components.
- Virtualization technology is used to create virtual machines.
- Managed entirely by the cloud provider.

2. Platform Layer

- Provides operating systems and runtime environments.
- Includes middleware, application servers, and database management systems.
- Supports programming languages and frameworks.
- Offers development tools such as SDKs and APIs.

3. Application Layer

- Contains user-developed applications.
- Applications are built, tested, deployed, and managed on the provided platform.

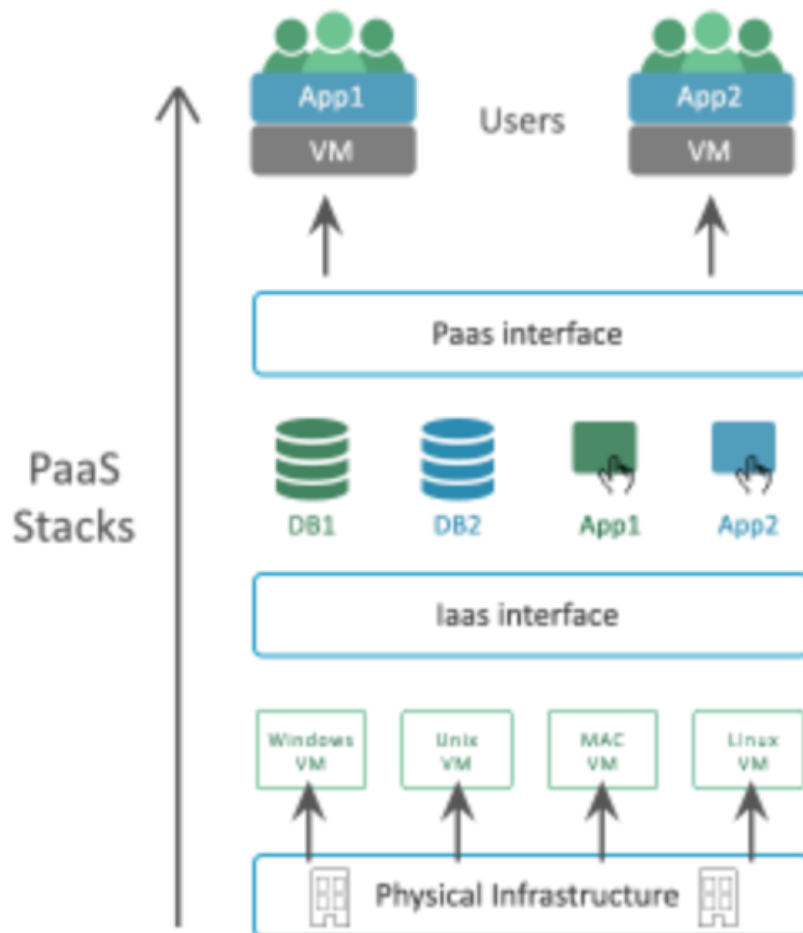
Characteristics of Platform as a Service (PaaS)

- Managed Infrastructure – The cloud provider manages servers, storage, networking, and virtualization.
- Integrated Development Environment – Provides built-in tools for application development, testing, and deployment.
- Multi-Tenancy – Multiple users share the same platform resources securely.
- Automatic Scaling – Resources are allocated automatically based on application demand.
- Support for Multiple Languages – Supports various programming languages and frameworks.
- Pay-Per-Use Model – Users are charged based on resource usage.

Case Study: Google App Engine

Google App Engine provides a platform to develop and deploy web applications without managing servers.

- Developers upload code to the platform.
- Google manages servers, operating systems, and scaling.
- Built-in services such as databases and authentication are provided.
- Applications automatically scale based on traffic.



15. With an intricate diagram Justify the cloud delivery models

Cloud delivery models define how computing services are provided to users over a network. The three primary cloud delivery models are:

- 1) Infrastructure as a Service
- 2) Platform as a Service
- 3) Software as a Service

1. Infrastructure as a Service (IaaS)

Infrastructure as a Service provides virtualized computing resources over the internet. Users rent infrastructure components such as virtual machines, storage, and networks.

Characteristics

Provides virtualized hardware resources
Users manage operating systems and applications
Pay-as-you-use billing model

IaaS is justified when organizations require full control over the operating system and software environment without investing in physical hardware. It reduces capital expenditure and offers flexibility in resource allocation.

Examples:

Amazon EC2
Microsoft Azure Virtual Machines
Google Compute Engine

2. Platform as a Service (PaaS)

Platform as a Service provides a development and deployment environment in the cloud. It includes runtime, middleware, and operating systems.

Characteristics

Developers focus only on application development
Infrastructure management is handled by the provider
Built-in development tools and database services

PaaS is justified for application developers who do not want to manage infrastructure. It improves productivity by providing a ready-to-use development platform.

Examples:

Google App Engine
Microsoft Azure App Service
Heroku

3. Software as a Service (SaaS)

Software as a Service delivers complete applications over the internet. Users access software through web browsers without installation.

Characteristics :

Fully managed by service provider
Accessible through internet
Subscription-based pricing

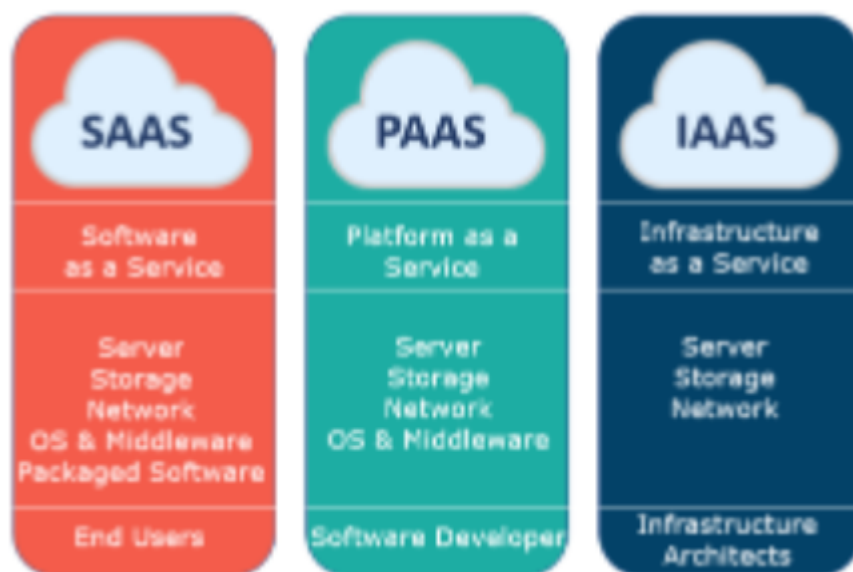
SaaS is justified for end users who require ready-to-use applications without maintenance responsibility. It eliminates installation, updates, and infrastructure management.

Examples:

Gmail

Microsoft 365

Salesforce



16. What are different types of clouds. Justify each type with a diagram and example.

Types of Clouds in Cloud Computing:

Clouds are classified based on their deployment model, which defines ownership, access control, and infrastructure location. The main types of clouds are:

- * Public Cloud
- * Private Cloud
- * Community Cloud
- * Hybrid Cloud

1. Public Cloud

A Public Cloud is a cloud infrastructure owned and operated by a third-party service provider. Resources are delivered over the internet and shared among multiple users.

Characteristics

- Infrastructure owned by service provider
- Available to the general public
- Pay-per-use model
- Multi-tenant environment

Advantages

- No capital investment
- Easy scalability
- High availability

Disadvantages

- Limited control over infrastructure
- Security concerns for sensitive data

Examples

- Amazon Web Services
- Microsoft Azure
- Google Cloud Platform

2. Private Cloud

A Private Cloud is a cloud infrastructure dedicated to a single organization. It may be hosted on-premises or by a third party.

Characteristics

- Single organization access
- Greater control and customization
- Enhanced security
- Can be on-site or externally hosted

Advantages

- Improved data security
- Better compliance management
- Custom configuration

Disadvantages

- High setup and maintenance cost
- Requires skilled IT staff

3. Community Cloud

A Community Cloud is shared by multiple organizations that have common objectives such as regulatory requirements, security policies, or industry standards.

Characteristics

- Shared among related organizations
- Managed internally or by third party
- Cost distributed among participants

Advantages

- Cost sharing
- Industry-specific compliance support

Disadvantages

- Limited scalability compared to public cloud
- Governance complexity

4. Hybrid Cloud

A Hybrid Cloud is a combination of two or more cloud types, typically public and private, connected through secure integration.

Characteristics

- Workload distribution between environments
- Data and application portability
- Flexible deployment

Advantages

- Better flexibility
- Improved cost efficiency
- Suitable for dynamic workloads

Disadvantages

- Complex management
- Integration challenges

